

TrafficVision AI:

Team Work & Individual Contribution Report

(Comprehensive Breakdown of Individual Responsibilities across Milestones 1 to 4)

TEAM 1

TEAM MEMBERS

1.Ch.Ushas Shri

2.Amit Halder

3.Asvitha J

1. Individual Contribution: Ch. Ushas Shri

Milestone 1 Week 1 & 2 : Project Establishing & System Design

- Defined key project objectives for TrafficVision AI, establishing scope across traffic monitoring, congestion management, prediction, alerts, and smart traffic workflows.
- Designed the Entity-Relationship (ER) diagram representing all core database entities and their relational attributes using Draw.io
- Created attributes where each entity contains :
 - UserID,Name,Email,Password,Role.

Road

-Road_ID,Road_Name,Location

Traffic Data

-Traffic_ID,Vehicle_Count,Average_Speed,Congestion_Level,Date_Time.

Milestone 2 Week 3 & 4: Congestion Forecasting Workflow

- Implemented the congestion forecasting workflow as part of the core AI/ML traffic prediction framework.
- Successfully integrated the congestion prediction workflow into the overall TrafficVision AI application pipeline.
- Formulated a weighted congestion scoring formula combining traffic density (60% weight) and speed factor (40% weight).
- Classified real-time traffic condition into three distinct levels: LOW, MEDIUM, and HIGH congestion.
- Configured automated system actions for each level, including triggering alerts and indicating alternate routes during HIGH congestion events.

Milestone 3 Week 5 & 6: Alert & Notification System Lifecycle

- Implemented the complete traffic alert workflow, including alert creation, state transitions, and status management.
- Established the full alert lifecycle with state tags: Active, Acknowledged, Notified, and Resolved.
- Connected traffic alerts directly with congestion events so that real-time incidents trigger corresponding system alerts.
- Developed congestion and accident notification mechanisms and integrated their representation within the dashboard.
- Validated notification workflows using simulated traffic/graphic incidents to verify correct alert generation and handling.

Milestone 4 Week 7 & 8: Testing, Integration & GitHub Version Control & Complete Documentation of The Project (Project, Individualwork, PPT)

- Configured and executed local development environments for both the React/Vite frontend and Node.js/Express backend in VS Code.
- Diagnosed and resolved critical frontend-backend CORS integration issues, enabling port 5175 (frontend) to communicate seamlessly with the Express backend.
- Executed end-to-end testing for alert management APIs.
- Verified the complete Active → Acknowledged → Resolved alert lifecycle with real-time updates reflected in PostgreSQL.
- Committed final implementation to Git (commit 5f9faed) ensuring .env security configuration was properly ignored.
- Updated the **main project documentation** with the implemented system details, AI/ML workflow, database design, testing results, security, deployment, challenges, future scope, and conclusion.
- Prepared the **project presentation** (PPT) covering the project architecture, implemented modules, results, testing, and overall project outcome.
- Ensured the **documentation** and presentation accurately reflect **individual contributions** as well as the overall TrafficVision AI project.

2. Individual Contribution: Amit Halder

Milestone 1 Week 1 & 2 :Environment Setup, DB Schema & Authentication

- Configured frontend environment using Vite + React (committed as main branch).
- Configured backend environment using Express + PostgreSQL (committed as backend-setup branch).
- Architected database schema containing 7 tables: users, locations, traffic_data, predictions, routes, route_locations, and alerts; verified in pgAdmin.
- Pushed relational database schema script (database/trafficvision_schema.sql) to GitHub repo.
- Established Express ↔ PostgreSQL database connection and verified via /test-db endpoint.
- Built user authentication system with password hashing (bcrypt) and JWT token creation for signup/login routes.

Milestone 2 Week 3 & 4: AI/ML Congestion Modeling & Travel Time Estimation

- Trained baseline Random Forest congestion classification model on real Bangalore traffic datasets.
- Mitigated model overfitting via hyperparameter regularization, attaining 89.3% test accuracy and a macro F1 score of 0.79.
- Developed Travel Time Estimation feature based on traffic density and road segment parameters.
- Generated automated Traffic Prediction Reports incorporating ML performance metrics.

Milestone 3 Week 5 & 6: Frontend Live Map, Analytics Dashboard & AI Recommendations

- Developed React (Vite) frontend featuring interactive live map with color-coded congestion markers and high-density heat circles.
- Built auto-refreshing disruptions/alerts panel and analytics dashboard tracking trends, busiest locations, congested routes, and alert stats with timeframe filters.
- Implemented Route Inspector displaying congestion scores, estimated travel times, and segment speeds.
- Engineered AI-Based Recommendation Engine analyzing historical + real-time bottleneck patterns to suggest alternate routes (e.g. suggesting Path B over Path A during spikes) and displaying time saved metrics.
- Created automated traffic performance summary reports highlighting congestion trends, incident counts, road performance, and actionable AI insights.

Milestone 4 Week 7 & 8: UI Responsiveness & System Optimization

- Frontend Performance: Optimized React components using React.memo, useMemo, and useCallback to reduce unnecessary re-renders.
- Map & Polling: Improved Leaflet map performance and introduced 30-second smart polling with tab-aware updates.
- Responsive Design: Added responsive support down to 320px and shimmer loading states for key dashboard components.
- Build & Database Optimization: Added database indexes and React code-splitting, reducing the initial bundle from 1,017 kB to 234 kB (77% reduction). Final validation achieved 47/47 Vitest tests passing.
- Testing & Validation: Improved overall application stability and verified the optimized frontend through comprehensive component testing and a successful production build.

3. Individual Contribution: Asvitha J

Milestone 1 Week 1 & 2 : Project Objectives, Schema Design & Figma Wireframing

- Participated in defining key project objectives for the TrafficVision AI system.
- Contributed to database schema design covering core entities: users, locations, traffic data, predictions, routes, and alerts.
- Designed user interface wireframes and visual layouts using Figma to guide frontend development.

Milestone 2 Week 3 & 4: Rule-Based Congestion Forecasting Logic & Weighting Rules

- Designed congestion forecasting and notification logic incorporating traffic inputs: Vehicle Count, Road Capacity, Average Speed, and Speed Limit.
- Implemented rule-based congestion forecasting logic within congestionlogic.js.
- Formulated a weighted congestion scoring formula combining traffic density (60% weight) and speed factor (40% weight).
- Configured automated system actions for each level, including triggering alerts and indicating alternate routes during HIGH congestion events.

Milestone 3 Week 5 & 6: Backend & Application Support Engineering

- Configured Node.js and Express backend application environment.
- Connected Express backend with PostgreSQL using project database configuration files.
- Verified active database connection through the /test-db API route.
- Managed package installation and environment setup using npm install.
- Successfully started backend server and verified browser execution via 'TrafficVision AI backend is running' response.
- Prepared backend integration endpoints for congestion forecasting logic API.

4. Summary & Team Responsibilities Overview

The TrafficVision AI project is an end-to-end intelligent traffic monitoring, congestion forecasting, and alert management system. The development effort was divided across four key milestones covering system design, AI/ML modeling, frontend/backend engineering, and database setup. Below is a summary matrix outlining each team member's core responsibilities and deliverable scope.

Team Member	Primary Domain	Key Milestone Deliverables
Ch. Ushas Shri	Alert Systems, System Integration & Testing, Backend Setup	ER Diagram, Alert notification integration, React-Express CORS Resolution, Alert API Verification & GitHub Branch Commit
Amit Halder	ML Modeling, Full-Stack Architecture & DB Schema, Backend setup, UI responsiveness.	Database Schema (7 tables), Random Forest Congestion Model, Live Map & Analytics Dashboard, AI Route Recommendations
Asvitha J	Congestion Logic, UI Wireframing, Backend application	Figma Wireframes, Weighted Rule-Based Congestion Forecasting (congestionlogic.js), Node.js/Express

The project was developed through various stages, including planning, design, development, machine learning, ML application integration, testing, deployment, and documentation. Individual responsibilities were integrated into the shared GitHub repository and managed through a milestone-based development process.